# MANIFOLD, GAS CYLINDER
# (MANIFOLD ML-344/TMQ-3)

## 1. SCOPE

1.1 This specification covers a metal device by means of which four Hydrogen Generators ML-303/TM are assembled to a common outlet which serves as a nozzle for attachment and inflation of sounding balloons.

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| NN-B-591 | Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment) |
| NN-B-601 | Boxes, Wood-Cleated-Plywood (For Domestic Shipment) |
| NN-B-621 | Boxes, Wood, Nailed, and Lock-Corner |
| UU-T-111 | Tape; Paper, Gummed (Sealing and Securing) |

MILITARY

| | |
|---|---|
| JAN-P-105 | Packaging and Packing for Overseas Shipment - Boxes; Wood, Cleated, Plywood |
| JAN-P-106 | Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed |
| MIL-B-107 | Boxes, Wood, Wirebound (Overseas Type) |
| JAN-P-108 | Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior |
| MIL-P-116 | Preservation, Methods of |
| JAN-P-127 | Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistar |
| *MIL-H-11127 | Hydrogen Generator ML-303/TM |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection By Attributes |
| MIL-STD-129 | Marking of Shipments |

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DRAWINGS

SIGNAL CORPS

| | |
|---|---|
| *SC-D-7431 | Hydrogen Generator ML-303/TM, Assembly and Details |
| SC-DL-86460 | Manifold ML-344/TMQ-3, Drawing and Data List |

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Preproduction samples.-

3.1.1 Quantities.- After award of contract two preproduction samples of Manifold ML-344/TMQ-3 shall be furnished.

3.1.2 Compliance with specified requirements.-

3.1.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.1.2.3 for exception.)

3.1.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.1.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.1.2.3 When deviation from 3.1.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.1.3 Test data.- The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.1.4 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used in reference standards to resolve any differences of opinion regarding interpretation of requirements.

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3.2 Construction.- Manifold ML-344/TMQ-3 shall be constructed in accordance with this specification and Drawing SC-DL-86460.

3.3 Parts, materials, and processes.- Parts, materials, and processes shall conform to the applicable requirements of Drawing SC-DL-86460. Unspecified parts, materials, and processes shall be selected by the contractor, such that they will be entirely suitable for the intended purpose.

3.4 Welding.- Welds shall be of ample size. Wherever possible, the welded joint shall be such that grinding on the finished weld will be unnecessary. The welds shall meet the tests of 4.3.

3.4.1 Cleaning.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.4.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved.

3.5 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.6 Workmanship.- All parts shall be manufactured and assembled in a thoroughly workmanlike manner.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection, general.- Inspection shall consist of group A inspection as specified in 4.1.1. Samples for inspection shall be selected in a unbiased manner by the Government inspector.

4.1.1 Group A inspection.- Group A shall consist of the Group A inspection specified in Table I and shall be performed by either of the following methods or a combination of both:

(a) 100-percent inspection.

(b) Statistical sampling in accordance with Standard MIL-STD-105. The acceptable quality levels (AQL's) shall be as specified in Table I or the contract.

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4.1.2 Disposition of rejected product.- When rejected product (sample unit, lot, or other portion of product) is submitted for inspection, the product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.1.2.1 Group A.- Disposition of product rejected during group A inspection shall be in accordance with Standard MIL-STD-105.

Table I - Inspection procedures

| Inspection | Test Par. | Preproduction | Group A AQL Major | Group A AQL Minor |
|---|---|---|---|---|
| Visual and Mechanical | 4.2 | X | 1% | 4% |
| Welding | 4.3 | X | 1% | |

Note: The following unlisted preproduction inspection may be performed to determine compliance with specified requirements: Nondestructive inspection on preproduction samples of equipment, and destructive or nondestructive inspection on separate preproduction samples of parts or materials. (See 3.1.)

4.2 Visual and mechanical inspection.- Manifold ML-344/TMQ-3 shall be thoroughly inspected for compliance with the requirements of this specification and the subsidiary drawing.

4.3 Welding.- The welded connections shall be tested to withstand an air pressure of 10 psi as follows: An air hose shall be connected to the large steel tube and each of the four branches shall be sealed. The pressure shall be adjusted to 10 psi and the manifold immersed in water; there shall be no evidence of leakage around the welds.

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## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment.- Manifold ML-344/TMQ-3 shall be preserved and packaged in accordance with Method III of Specification MIL-P-116.

5.1.2 For export shipment.- Preserve and package Manifold ML-344/TMQ-3 in accordance with procedures specified for the designated methods as prescribed in Specification MIL-P-116, and further described in the following paragraphs.

5.1.2.1 Manifold ML-344/TMQ-3.- Package each item Method III, as follows: Cushion each item by wrapping in flexible corrugated paper. Secure the cushioning with gummed Kraft tape conformint to Specification UU-T-111.

5.1.2.2 Consolidated package.- Consolidate the items, packaged as specified in 5.1.2.1, together within a suitable style and grade fiberboard box, conforming to Specification JAN-P-108. Fill all voids with flexible corrugated paper to prevent movement. Seal entire box closure, corners and joints, with water-resistant pressure-sensitive tape, conforming to Specification JAN-P-127.

5-2 Packing.-

5.2.1 For domestic shipment.- Manifold ML-433/TMQ-3 shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, or corrugated fiberboard boxes conforming to Specification NN-B-591, NN-B-601, NN-B-621, and LLL-B-631, respectively. Closures shall be made in accordance with the applicable box specification.

5.2.2 For export shipment.- Place a suitable quantity of consolidated packages within a wood cleated plywood nailed wood or wire bound wood box conforming to the requirements of Specifications JAN-P-105, JAN-P-106 and MIL-B-107 respectively. Fabricate the boxes to fit contents snugly. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping container as specified in the appendix of the applicable box specification.

5.2.2.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.3 Marking for shipment.- Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129.

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether domestic or overseas shipment is required.

(c) Samples required. (See 3.1.)

(d) Marking and shipping of samples.

(e) Acceptable quality level. (See 4.1.1(b).)

6.2 Contractual requirements.- It is recommended that the following be included in the contract:

6.2.1 Additional testing.- It should be understood that additional inspection and non-destructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable document. The government inspector, therefore, may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection and testing, performed either at a Government laboratory or the contractor's plant.

6.2.2 Waiver of preproduction samples.- When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request should include a statement indicating that the specified samples have been submitted and approved on the current contract; production of the item under the new order will be concurrent with, or follow without interruption, production on the current order; and current production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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